

8	(a) (i) time for initial number of nuclei/activity to reduce to one half of its initial value	M1 A1 [2]
	(ii) $\lambda = \ln 2 / (24.8 \times 24 \times 3600)$ $= 3.23 \times 10^{-7} \text{ s}^{-1}$	M1 A0 [1]
	(b) (i) $A = \lambda N$ $3.76 \times 10^6 = 3.23 \times 10^{-7} \times N$ $N = 1.15 \times 10^{13}$	C1 A1 [2]
	(ii) $N = N_0 e^{-\lambda t}$ $= 1.15 \times 10^{13} \times \exp(-\{\ln 2 \times 30\} / 24.8)$ $= 4.97 \times 10^{12}$	C1 A1 [2]
	(c) ratio $= (4.97 \times 10^{12}) / (1.15 \times 10^{13} - 4.97 \times 10^{12})$ $= 0.76$	C1 A1 [2]

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Section B

0 (a) a reduced gain